



Name: _____ HR: _____

6th Grade Math Unit 5: Division of Fractions and Volume STAR Cards

5

QUESTION	ANSWER
5a) ☆☆☆ 1. What is a fraction? 2. What is the top called? → $\frac{3}{5}$ 3. What is the bottom called? ↗	5a) $\frac{\text{part}}{\text{whole}}$ 1. A fraction is a number representing a part of a whole 2. Numerator (how many equal parts you have) → $\frac{3}{5}$ 3. Denominator (how many equal parts make up the whole) ↗
5b) ☆☆☆ 1. How do we <u>simplify fractions</u>? 2. How can you simplify $\frac{6}{15}$?	5b) 1. Divide the numerator(top) and denominator(bottom) by the same number (common factor). 2. Divide 6 and 15 by <u>3</u> which is a <u>common factor</u>. $\frac{6}{15} \div 3 = \frac{2}{5}$

5c)



1. How do we convert a mixed number into an improper fraction?

Example: $4\frac{1}{3}$

2. How do we change an improper fraction into a mixed number?

Example:

$$\frac{10}{7}$$

5c)

1.

2nd) Add the numerator

$$4\frac{1}{3} = \frac{13}{3}$$

The diagram shows the conversion of the mixed number $4\frac{1}{3}$ to the improper fraction $\frac{13}{3}$. A curved arrow labeled '+ 1' points from the whole number 4 to the numerator 1. Another curved arrow labeled 'x 3' points from the whole number 4 to the denominator 3. The result is shown as $\frac{13}{3}$.

1st) Multiply the whole number by the denominator (=12)

3rd) Keep the same denominator

2. Divide the numerator by the denominator. Write your remainder as a fraction.

$$7 \overline{)10} = 1\frac{3}{7}$$

The diagram shows the division of 10 by 7. 7 goes into 10 one time, with a remainder of 3. The result is written as $1\frac{3}{7}$.

5d)



Explain the rules for multiplying fractions with this example:

$$\frac{2}{3} \times \frac{4}{5}$$

5d)

Multiply straight across

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

The diagram shows the multiplication of $\frac{2}{3}$ and $\frac{4}{5}$. Two horizontal arrows point from the numerators 2 and 4 to the numerator 8. Two horizontal arrows point from the denominators 3 and 5 to the denominator 15. The result is $\frac{8}{15}$.

5e)



How do we divide fractions?

Example:

$$\frac{1}{5} \div \frac{3}{4}$$

5e)

To divide fractions

$$\frac{1}{5} \div \frac{3}{4}$$

Keep Change Flip

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{1}{5} & \times & \frac{4}{3} \end{array}$$

*then multiply
straight across*

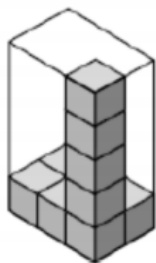
$$\frac{1}{5} \times \frac{4}{3} = \frac{4}{15}$$

5f)



1. What is volume?

2. How can you quickly find the volume of this figure?



5f)

1. Volume is the number of cubes it takes to fill up a 3D shape.

2. Step 1: Find the area of the bottom layer.

Step 2: Multiply the area of the bottom layer by the height.

OR use the formula $V = L \cdot W \cdot H$

$$V = 3 \cdot 2 \cdot 5$$

$$V = 30 \text{ cubes}$$

